



Stylish Fairphone 2

Launched last year in December, Fairphone 2 has been revamped with a new design along with other changes. <http://dgit.in/FairPh2>



Gear 360 available

Launched earlier this year, Samsung is now working to make it available across more offline stores. <http://dgit.in/SG360avail>

Industry Connect

sion of robots and certainly soon we're going to see a \$5000 desktop robot that will have the capabilities of a \$50k or \$75k machine of today.

There's a fourth kind of technology which is also affecting the future of making called compose. Compose is our ability to not only understand traditional material and existing material to in order to use them in better and more specific ways, but also to generate new classes of materials by selecting the properties you would like and then literally designing the matter or substance out of which the material is made from, to have the properties that you want or like.

d How is this process of creating customized materials carried out?

Wujec: It's done in a variety of ways. For example concrete is the world's most commonly used building material. We also know that concrete is brittle, it breaks. So wouldn't it be great to have concrete that knows when it's broken and then heals itself? An interesting way to do that, I think, was discovered recently in a microbe that was found at the bottom of the oceans in those volcanic vents. These microbes in the presence of water will manufacture silicates, taking existing material and literally make rock. The microbe can exist in extreme conditions. So the next generation of concrete has this microbe in there and it lies there dormant until it detects water. So if there's a crack in the concrete the water will get through and the microbe is activated.

d That's like living concrete. How are these materials made?

Wujec: It is, it is! We're increasingly seeing these examples of re-engineering materials into other classes of materials – something called metamaterials. Metamaterials are generally 3D printed and in the form of three dimensional micro lattice. Since printers don't care what you print, you can create very complicated internal structures that the eye can't even see that has properties that are really interesting. Take jello for example, when squeeze it,

it expands. But with a lattice material you program how it reacts. Helmets for example, brain concussions are a big issue in sports, you could have a helmet that is soft on the inside that contours exactly to your skull. The material gradually goes to hard, so it's like an extra bone casing.

d How can makers at the very grass-root level play into this entire grand vision?

Wujec: Makers are in a fantastic, incredible space in the history of humanity because for the first time in history, almost anyone who has a smartphone and an internet can get access to learning how to make things. There are incredible websites like Instructables that provide access and inspiration to design and make things. Secondly, many software tools are increasingly free to be able to design and make things.

I'm proud to work at a company that's so passionate about making things. Our CEO is himself a maker. He's made boats, he's built houses, a gokart which he's now turning into a self-driving go kart for his teenage boys. Many CEOs spend their money lavishly on villas and cars and Carl went and bought two maker spaces for himself.

d Many of our readers would be interested in knowing what a "Fellow" at Autodesk really does?

Wujec: My role at Autodesk involves work in the office of the Chief Technology Officer. Our overall group's goal is to do three things: explore, exploit, and explain the technologies that are important to the industries that we serve and the customers that we serve. And our horizon is the five to twenty year horizon and even beyond. So our group explores everything from synthetic biology, which I described, to robotic vision, to something called metadata visualization and systems thinking. We do something called parametric humans, which is literally a twenty year project to completely digitize the human, from the musculature to the nerves to the bones so you can run simulations on a digital human.

Since we produce things for people, if we create a Digital Human – which is a very difficult task – then that is yet another tool for designers to digitally test things before they physically produce them.

d What are your thoughts on digital consciousness or the self awareness of a machine as an individual?

Wujec: Well I think there are layers of consciousness and people have been debating the nature of consciousness for a very long time. There is what's called the easy and the hard question. Easy is the mimicry so absolutely, AI like Siri or Cortana can pass the Turing test. But does it have cognition in the sense of a felt reality?

So I tend to think that that's a systems model and that consciousness can be modeled through other types of perhaps non-biological systems. This is my own personal point of view not Autodesk's. I think it's going to be a long way away.

When it comes to machine learning, this is how it works. So a word, any word that is parsed whether by Google, or Siri, or Cortana has a number of numbers associated to it through statistical analysis. So the word 'king' for example, might have 30,000 numbers that are generated through statistical analysis. The computer represents it as a vector, and that means it's a line in space at a certain distance in 30,000 dimensions, so not one, not two, not three, but 30,000. Every other word has its own dimensionality and then you can perform operations on these words to create meaning. So for example if you take the word 'king', that's 30,000 numbers, and then you subtract the word 'man', so that's another word, and you end up with another number and that number ends up corresponding to the word 'queen'. So we both intuitively know, semantically how they're related. But the computer does it only statistically, it doesn't have that felt sense of, 'ohh, I get it' like we all experience. **d**

To read the full interview:
<http://dgit.in/AutoDsknt>